Where we stand today

The ML stack is built and integrated — the product layer that makes it CreditMitra is the remaining work

50%

of the overall project is complete

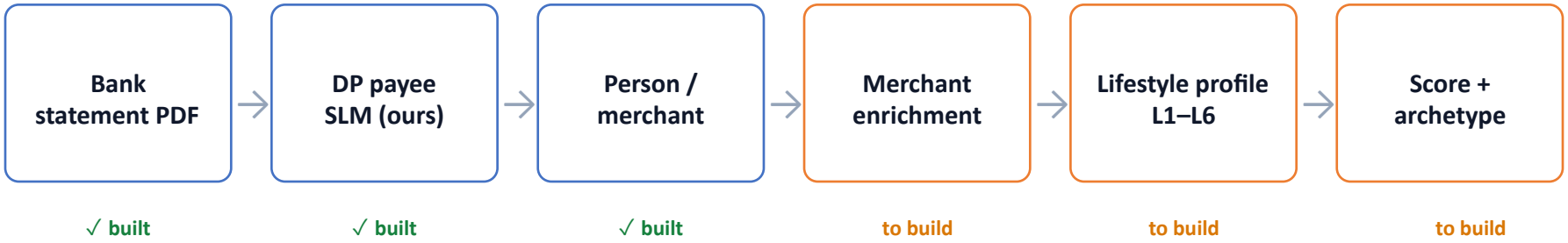
0%

of the core-IP lifestyle + scoring layer has been started that is this plan

- **DP payee-extraction SLM** — DONE
  Qwen2.5-1.5B + QLoRA + DP-SGD , F1 84% , ε≈2 ,MIA AUC≈0.5 ·
- **Merchant / person classifier** — DONE
  ONNX Qwen3-Embedding → AttentionMLP, integrated as a singleton service
- **Engine ingest pipeline** — DONE
  PDF → Docling tables → payee-lora via Ollama → classification → task store + API
- **Dashboard shell** — DONE
  Upload, live processing status, stats, enriched transaction ledger
- **Lifestyle profile + scorecard + risk UI** — THIS PLAN
  L1–L6 indices, archetypes, two-block score, enrichment server, report page

The product we are building

A bank statement becomes a merchant-derived lifestyle profile computed on-device, privately with an explainable score as its output



Thesis: merchant-derived lifestyle profiling for explainable, privacy-preserving credit assessment of India's thin-file borrowers. The lifestyle layer is only possible because our DP model resolves raw UPI narrations to merchants that combination is the contribution.

Contract-first, three parallel tracks

Five Pydantic schemas written once, merged to main, then frozen:

- **Transaction (extended)**
- **MerchantEnrichment**
- **FeatureVector**
- **LifestyleProfile**
- **CreditRiskReport + ScoreFactor**

Track A · Data pipeline

Amount/date parsing, ~300-merchant India dictionary, enrichment module + SQLite cache, pipeline wiring, standalone merchant-server

codes against real PDFs

Track B · Risk brain

Feature engineering, L1–L6 lifestyle profile + archetypes, two-block scorecard, risk API, persona validation

codes against a frozen JSON fixture

Track C · Frontend + integration

TS types + mock API, Lifestyle Report page (the demo), analytics charts, table upgrade, end-to-end demo

codes against sample_report.json

Only two sync points: Checkpoint 1 (A wires B's pure functions) and Checkpoint 2 (C swaps mock → real API). Single-owner file map keeps merge conflicts near zero.

Six lifestyle indices

L1 Essential Stability

Essential-category share × month-to-month spending persistence (cosine similarity)

L2 Aspirational

Discretionary lifestyle weight — food delivery, entertainment, travel, quick -commerce

L3 Digital Maturity

(1 − cash ratio) × merchant-resolution rate — a signal unique to our pipeline

L4 Commitment

Voluntary recurring obligations (rent, SIP, insurance) × day-of-month regularity

L5 Leverage (inverse)

BNPL dependence , Simpl, LazyPay, slice : hidden leverage visible before any bureau

L6 Risk Appetite (inverse)

Gambling + crypto share with a steep 10 × penalty curve, per Indian underwriting practice

Referenced from: framework follows Gladstone et al. (EPJ Data Science 2021, 74M transactions) persistence, burstiness, category diversity and turnover feed the indices as behavioral texture.

Archetypes + a two-block explainable scorecard

Six rule-based archetypes

Disciplined Urban Salaried :prime thin-file candidate

Saver / Investor :excellent

Aspirational Spender :lend with limits

Gig Hustler :income-volatile, not undisciplined **Credit-Dependent :**high risk

Cash-Economy Earner :refer to manual review

Gig-payout detection (Zomato as a payer) identifies livelihood — no generic cashflow model can.

Cash-flow block · ~±300 pts

FOIR, salary regularity, balance buffer, bounces, overdrafts ,the literature-backed baseline (NBER cash-flow underwriting).

Lifestyle block · ~±300 pts

Points from L1–L6, each emitted as a named factor: “Commitment 78 , sustained SIP + insurance + rent → +45”.

300–900 band retained. Every point traceable to a factor; healthcare excluded (fair lending).

Two-tier enrichment behind a hard privacy boundary

Everything sensitive stays on-device; only bare merchant names ever cross the wire

ON-DEVICE (never leaves)

- Raw PDF, narrations, balances, all PII
- DP payee SLM (ε≈2 formal guarantee) via Ollama
- Person/merchant classifier (ONNX, CPU)
- Feature engineering, L1–L6 profile, score
- Local merchant dictionary (~300 India merchants, 6 fields) as offline fallback

→

only ["Swiggy", "Jio", …]

Merchant server

- Lookup order: SQLite cache → India dictionary → Langchain fallback (taxonomy-constrained) → cache
- Returns the 6-field enrichment contract: category, essential, risk flag, lifestyle dim, recurring type
- Standalone FastAPI repo, API-key auth, Dockerised

Acceptance test: a privacy-payload test asserts that only merchant name strings appear in outbound requests , no narrations, amounts, dates or personal payees, ever.

Validation without labeled defaults

disciplined_saver | stable_salaried | gig_irregular | cash_heavy | over_leveraged | frequent_bouncer

Assertion 1: archetype mapping

Each persona must be assigned its intended archetype by the rule engine (e.g., gig payout credits → Gig Hustler, not Frequent Bouncer).

Assertion 2 : score ranking

disciplined_saver > stable_salaried > … > over_leveraged / frequent_bouncer. Weights in scoring_config.py are tuned unt il this ranking holds — that tuning is the acceptance test.

Assertion 3 : EPJ-consistency checks

Sanity checks against the literature: a bursty, irregular fixture must score lower L4 (Commitment) than a regular one; recurr ing rent + SIP must lift L4.

References

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